Excerpt from: BUP_Lost_and_Found_SRS (1).pdf

4. Non-Functional Requirements

4.1 Performance

The system shall remain responsive under expected campus-scale usage.

ID	Requirement	Priority
NFR-PERF-1	The system shall return API responses for standard listing requests (lost items, found items, categories) within 1 second under normal load.	Medium
NFR-PERF-2	The system shall support at least 500 registered student accounts and hundreds of concurrent active item posts without noticeable degradation.	Medium
NFR-PERF-3	The matching algorithm shall compute scores for a full category (dozens of items) in under 2 seconds.	Low

4.2 Security

ID	Requirement	Priority
NFR-SEC-1	The system shall store all passwords as bcrypt hashes with a minimum cost factor of 10; plaintext passwords shall never be persisted or logged.	High
NFR-SEC-2	The system shall use signed JSON Web Tokens for session authentication, with a 12-hour expiry.	High
NFR-SEC-3	The system shall restrict administrator-only endpoints (category management, match confirmation) using server-side role verification, not merely client-side hiding.	High
NFR-SEC-4	The system shall never expose a found item's finder identity through any public (unauthenticated) endpoint.	High
NFR-SEC-5	The system shall validate and sanitize all user-submitted input on the server side before persisting it to the database.	Medium

4.3 Usability

ID	Requirement	Priority
NFR-USE-1	The system shall present a single, uncluttered landing page rather than a divided multi-panel layout.	High
NFR-USE-2	The system shall provide type-ahead suggestions for all name, category, location, size, and color input fields to minimize typing effort and input errors.	High
NFR-USE-3	The system shall use a body text size and layout suitable for comfortable reading on both desktop and mobile screens.	Medium
NFR-USE-4	The system shall not force a password change immediately after login, preserving a smooth first-time user experience.	Medium
NFR-USE-5	The interface shall follow a consistent visual theme (color palette, typography) across all pages.	Low

4.4 Reliability & Availability

ID	Requirement	Priority
NFR-REL-1	The system shall handle database connection failures gracefully, returning a clear error rather than crashing the server process.	Medium
NFR-REL-2	The system shall use database transactions when confirming a match, ensuring both the lost and found records update atomically.	High
NFR-REL-3	The system should target at least 99% uptime during university operating hours once deployed.	Low

4.5 Maintainability

ID	Requirement	Priority
NFR-MAINT-1	The codebase shall separate concerns across routes, middleware, and utility modules on the backend, and pages/components on the frontend.	Medium

ID	Requirement	Priority
NFR-MAINT-2	Reference data (categories, locations, sizes, colors, departments) shall be centrally defined in a single module to simplify future updates.	Medium
NFR-MAINT-3	The system shall include a seed script capable of repeatably generating consistent sample data for development and testing.	Low

4.6 Scalability

ID	Requirement	Priority
NFR-SCALE-1	The database schema shall support growth in the number of students, categories, and item posts without structural redesign.	Medium
NFR-SCALE-2	The system architecture shall allow the frontend and backend to be deployed and scaled independently.	Low

4.7 Portability

ID	Requirement	Priority
NFR-PORT-1	The frontend shall run correctly on current versions of major browsers (Chrome, Firefox, Edge, Safari).	Medium
NFR-PORT-2	The backend shall run on any standard Node.js hosting environment without dependence on a specific operating system.	Low

4.8 Data Privacy

ID	Requirement	Priority
NFR-PRIV-1	The system shall withhold a found item's finder identity from all users except the administrator and the finder themselves.	High
NFR-PRIV-2	The public student directory shall exclude phone numbers and other sensitive contact information.	Medium
NFR-PRIV-3	Lost item posts, which intentionally reveal the owner's identity to aid recovery, shall clearly convey this to the student at the time of posting.	Medium